



Storm Intensity Distribution — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 20-March-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs (“MKM”).

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an “as is” basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

1 Test Results — Storm Intensity Distribution (MKM-SI-001)

Tests run on **20 March 2026 at 12:15:19**.

Table 1: Test Results Summary — Storm Intensity Distribution (MKM-SI-001)

Total Tests	105
Passed	105
Failed	0
Skipped	0
Duration	0.24s

Table 2: Detailed Test Results — Storm Intensity Distribution

Test Description	Acceptance Criteria	Result	Time
Custom parameters	d.params.base_mean == 50;	Pass	0.000s
	d.params.tail_threshold == 70		
Default parameters	d.params.base_mean == 45.0;	Pass	0.000s
	d.params.base_std == 15.0 (+4 more)		
From params object	d.params.base_mean == 55	Pass	0.000s
From scenario baseline	d is not None; d.params.name == "baseline"	Pass	0.000s
From scenario unknown raises	—	Pass	0.000s
Seed gives deterministic output	s1 == s2	Pass	0.000s
Tail prob between 0 and 1	0.0 < d._tail_prob < 1.0	Pass	0.000s
All categories defined	set(DURATION_PARAMS.keys()) == expected	Pass	0.000s
Catastrophic extended	high == 240	Pass	0.000s
Each tier’s max roughly doubles the previous.	1.5 <= ratio <= 3.0	Pass	0.000s
Extreme extended	DURATION_PARAMS["extreme"] == (36, 72, 144)	Pass	0.000s
Moderate extended	low == 12; mode == 30 (+1 more)	Pass	0.000s
Severe extended	DURATION_PARAMS["severe"] == (24, 48, 96)	Pass	0.000s
Gap types	GapType.SHORT.value == "short"; GapType.MEDIUM.value == "medium" (+1 more)	Pass	0.000s
Sequence gap mapping	SEQUENCE_GAP_TYPE[SequenceType.DOUBLET] == GapType.MEDIUM; SEQUENCE_GAP_TYPE[SequenceType.CLUSTER] == GapType.MEDIUM (+2 more)	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Sequence types	SequenceType.ISOLATED.value == "isolated"; SequenceType.DOUBLET.value == "doublet" (+2 more)	Pass	0.000s
Storm count ranges	SEQUENCE_TYPE.STORM.COUNTS[SequenceType.ISOLATED] == (1, 1); SEQUENCE_TYPE.STORM.COUNTS[SequenceType.DOUBLET] == (2, 2) (+2 more)	Pass	0.000s
Above max is zero	d.exceedance_probability(d.params.max_intensity + 10) == 0.0	Pass	0.000s
Above threshold uses pareto	0 < p < 1	Pass	0.000s
At max intensity is zero	d.exceedance_probability(d.params.max_intensity) == 0.0	Pass	0.000s
At min intensity is one	d.exceedance_probability(d.params.min_intensity) == 1.0	Pass	0.000s
At threshold continuous	abs(p_just_below - p_just_above) < 0.05	Pass	0.000s
Below min is one	d.exceedance_probability(d.params.min_intensity - 5) == 1.0	Pass	0.000s
Below threshold uses normal	0 < p < 1	Pass	0.000s
Monotonically decreasing	all(probs[i] > probs[i + 1] for i in range(len(probs) - 1))	Pass	0.000s
Constants	EVENT_WINDOW_HOURS == 168; MIN_DRAINAGE_WINDOW_HOURS == 12 (+1 more)	Pass	0.000s
Exact boundary	fits_in_window([78.0, 78.0], [0.0])	Pass	0.000s
Exceeds window	not fits_in_window([72.0, 72.0], [48.0])	Pass	0.000s
Simple doublet fits	fits_in_window([24.0, 24.0], [48.0])	Pass	0.000s
Single storm	fits_in_window([100.0], [])	Pass	0.000s
Long gaps	GAP_PARAMS[GapType.LONG] == (48, 96, 144)	Pass	0.000s
Medium gaps	GAP_PARAMS[GapType.MEDIUM] == (24, 48, 72)	Pass	0.000s
Short gaps	GAP_PARAMS[GapType.SHORT] == (6, 18, 36)	Pass	0.000s
Doublet medium	get_gap_range(SequenceType.DOUBLET) == (24, 48, 72)	Pass	0.000s
Isolated zero	get_gap_range(SequenceType.ISOLATED) == (0, 0, 0)	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Get duration range	<code>get_duration_range("moderate") == (12, 30, 60)</code>	Pass	0.000s
Get max duration	<code>get_max_duration("catastrophic") == 240; get_max_duration("minimal") == 12</code>	Pass	0.000s
Ids unique	<code>len(ids) == 100</code>	Pass	0.000s
Sequence id format	<code>sid.startswith("STORM-"); len(sid) == 14</code>	Pass	0.000s
Sequence ids unique	<code>len(ids) == 100</code>	Pass	0.000s
Storm id format	<code>sid.startswith("STORM-"); len(sid) == 14</code>	Pass	0.000s
One prob returns min	<code>d.inverse_exceedance(1) == d.params.min_intensity</code>	Pass	0.000s
Result in bounds	<code>d.params.min_intensity <= v <= d.params.max_intensity; d.params.min_intensity <= v <= d.params.max_intensity</code>	Pass	0.000s
Roundtrip above threshold	<code>abs(recovered - target) < 3.0</code>	Pass	0.000s
Roundtrip below threshold	<code>abs(recovered - target) < 2.0</code>	Pass	0.000s
Zero prob returns max	<code>d.inverse_exceedance(0) == d.params.max_intensity</code>	Pass	0.000s
Fat tail higher exceedance	<code>p_fat > p_thin</code>	Pass	0.000s
Pareto above threshold between 0 and 1	<code>0 < p < 1</code>	Pass	0.000s
Pareto at threshold is one	<code>d.pareto_exceedance(d.params.tail_threshold) == 1.0</code>	Pass	0.000s
Pareto below threshold is one	<code>d.pareto_exceedance(d.params.tail_threshold - 5) == 1.0</code>	Pass	0.000s
100yr greater than 10yr	<code>i100 > i10</code>	Pass	0.000s
Ordering 10 50 100	<code>i10 < i50 < i100</code>	Pass	0.000s
Result in bounds	<code>d.params.min_intensity <= v <= d.params.max_intensity; d.params.min_intensity <= v <= d.params.max_intensity</code>	Pass	0.000s
Storms per year affects result	<code>many > few</code>	Pass	0.000s
Invalid category	—	Pass	0.000s
Mode should be near the most frequent value in a large sample.	<code>abs(median - mode) < 15</code>	Pass	0.006s
Line 50: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally.	<code>low <= d <= high</code>	Pass	0.000s
Reproducibility	<code>d1 == d2</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All sampled durations fall within [min, max] for the category. [category='baseline']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='catastrophic']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='extreme']	<code>low <= d <= high</code>	Pass	0.001s
All sampled durations fall within [min, max] for the category. [category='minimal']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='moderate']	<code>low <= d <= high</code>	Pass	0.000s
All sampled durations fall within [min, max] for the category. [category='severe']	<code>low <= d <= high</code>	Pass	0.000s
Doublet gaps should fall in medium range (24-72h).	<code>all(24 <= g <= 72 for g in gaps)</code>	Pass	0.000s
Isolated returns zero	<code>sample_gap(SequenceType.ISOLATED) == 0.0</code>	Pass	0.000s
Line 49: <code>rng=None</code> → creates <code>np.random.RandomState()</code> internally.	<code>low <= g <= high</code>	Pass	0.000s
Persistent gaps should fall in short range (6-36h).	<code>all(6 <= g <= 36 for g in gaps)</code>	Pass	0.000s
Reproducibility	<code>g1 == g2</code>	Pass	0.000s
Within bounds[sequencetype.cluster] [seq_type=iSequenceType.CLUSTER: 'cluster']	<code>low <= g <= high</code>	Pass	0.000s
Within bounds[sequencetype.doublet] [seq_type=iSequenceType.DOUBLET: 'doublet']	<code>low <= g <= high</code>	Pass	0.000s
Within bounds[sequencetype.persistent] [seq_type=iSequenceType.PERSISTENT: 'persistent']	<code>low <= g <= high</code>	Pass	0.000s
Different seeds different output	<code>d1.sample(10) != d2.sample(10)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Sample n returns correct length	<code>len(samples) == 100</code>	Pass	0.000s
Sample single	<code>d.params.min_intensity <= v <=</code> <code>d.params.max_intensity</code>	Pass	0.000s
Samples within bounds	<code>all(d.params.min_intensity <= s <=</code> <code>d.params.max_intensity...</code>	Pass	0.001s
Seeded reproducibility	<code>s1 == s2</code>	Pass	0.000s
Create basic	<code>storm.storm_id == "STORM-abc12345";</code> <code>storm.duration.hours == 24.0</code>	Pass	0.000s
From dict round trip	<code>restored.storm_id ==</code> <code>original.storm_id;</code> <code>restored.duration.hours ==</code> <code>original.duration.hours (+6 more)</code>	Pass	0.000s
Precipitation rounding	<code>d["precipitation_mm"] == 35.12;</code> <code>d["intensity_factor"] == 1.234568 (+1</code> <code>more)</code>	Pass	0.000s
To dict	<code>d["storm_id"] == "STORM-test1234";</code> <code>d["start_time.hours"] == 48.0 (+6</code> <code>more)</code>	Pass	0.000s
Describe returns string	<code>isinstance(desc, str); "base_mean" in</code> <code>desc.lower() or "Base mean" in desc</code>	Pass	0.074s
Params roundtrip	<code>recovered.base_mean == 52;</code> <code>recovered.name == "custom"</code>	Pass	0.000s
Tail probability matches internal	<code>d.to_dict()['tail_probability'] ==</code> <code>pytest.approx(d._tail...</code>	Pass	0.000s
To dict keys	<code>'parameters' in result;</code> <code>'tail_probability' in result</code>	Pass	0.000s
Antecedent defaults	<code>seq.antecedent.soil_moisture ==</code> <code>"normal"; seq.antecedent.groundwater</code> <code>== "normal"</code>	Pass	0.000s
Create doublet	<code>seq.num_storms == 2; len(seq.storms)</code> <code>== 2 (+2 more)</code>	Pass	0.000s
From dict round trip	<code>restored.storm_id ==</code> <code>original.storm_id;</code> <code>restored.duration.hours ==</code> <code>original.duration.hours (+6 more)</code>	Pass	0.000s
Isolated no gaps	<code>len(seq.inter_storm_gaps_hours) == 0</code>	Pass	0.000s
Spatial correlation defaults	<code>seq.spatial_correlation_enabled</code> <code>is False;</code> <code>seq.spatial_correlation_range_km</code> <code>is None</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Spatial correlation enabled	<code>d["spatial.correlation.enabled"] is True; d["spatial.correlation.range_km"] == 43.2</code>	Pass	0.000s
To dict	<code>d["storm_id"] == "STORM-test1234"; d["start_time.hours"] == 48.0</code> (+6 more)	Pass	0.000s
Exceedance probs decreasing	<code>stats['prob.exceed_60'] >= stats['prob.exceed_70'] >= ...</code>	Pass	0.000s
Fat tail higher p99 than thin	<code>fat['p99'] > thin['p99']</code>	Pass	0.148s
Mean in reasonable range	<code>30 < stats['mean'] < 70</code>	Pass	0.000s
Percentiles ordered	<code>stats['p10'] <= stats['p25'] <= stats['p50'] <= \{\}</code> ...	Pass	0.000s
Required keys present	<code>k in stats</code>	Pass	0.000s
Duration mismatch	<code>any("total.duration.hours" in e for e in errors)</code>	Pass	0.000s
Exceeds precip window	<code>any("MAX_PRECIP_END_HOUR" in e for e in errors)</code>	Pass	0.000s
Invalid intensity category	<code>any("invalid intensity" in e for e in errors)</code>	Pass	0.000s
Invalid sequence type	<code>any("Invalid sequence.type" in e for e in errors)</code>	Pass	0.000s
Overlapping storms	<code>any("starts at" in e and "before storm" in e for e in err...</code>	Pass	0.000s
Storm count mismatch	<code>any("num.storms" in e for e in errors)</code>	Pass	0.000s
Too many storms	<code>any("not in [1, 5]" in e for e in errors)</code>	Pass	0.000s
Valid doublet	<code>errors == []</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (12 tests)	D. Kelly
07-Mar-2026	1.1	73 test(s) added; 12 test(s) removed (total: 146)	D. Kelly
08-Mar-2026	1.2	42 test(s) added (total: 115)	D. Kelly
12-Mar-2026	1.3	2 test(s) added (total: 117)	D. Kelly
20-Mar-2026	1.4	12 test(s) removed (total: 105)	D. Kelly